Staff mark scheme · questions 1 and 2

One mark for each listed point unless stated. Accept scientifically equivalent wording and agreed response modes.

1. 2 marks

1. Interphase.

2. So each daughter nucleus/cell receives a complete matching set of genetic information.

2. 2 marks

1. C before A before D.

2. B last, with the complete sequence C A D B for full credit.

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Staff mark scheme · questions 3 and 4

One mark for each listed point unless stated. Accept scientifically equivalent wording and agreed response modes.

3. 2 marks

1. 8 chromosomes in each daughter.

2. Chromosome copies separate so each daughter receives one complete set; the number is conserved.

4. 2 marks

1. Division increases cell number.

2. Elongation increases the length/size of cells.

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Staff mark scheme · questions 5 and 6

One mark for each listed point unless stated. Accept scientifically equivalent wording and agreed response modes.

5. 4 marks

1. Treatment: 9/15 × 100 = 60%.

2. Comparison: 3/15 × 100 = 20%.

3. Improvement is 40 percentage points higher in the treatment group; accept a correct equivalent numerical comparison.

4. A relevant limit: small groups, short follow-up, no adverse-event data given, or unclear allocation/definition of improvement.

6. 3 marks

1. DNA.

2. Section/sequence of DNA (containing information for a protein).

3. All the DNA of an organism.

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Staff mark scheme · questions 7 and 8

One mark for each listed point unless stated. Accept scientifically equivalent wording and agreed response modes.

7. 2 marks

1. T A G C.

2. Double helix.

8. 3 marks

1. Correct gamete labels: A and a for one parent; a and a (or just a in a valid 2×1 square) for the other.

2. Correct combinations: Aa, Aa, aa, aa, or equivalent two equally weighted combinations.

3. Probability aa = 1/2, 50%, or 2/4.

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Staff mark scheme · questions 9 and 10

One mark for each listed point unless stated. Accept scientifically equivalent wording and agreed response modes.

9. 2 marks

1. 50% (1/2).

2. Each fertilisation is independent; previous offspring do not change or use up the predicted chance.

10. 2 marks

1. The exact prediction is not justified / exceeds the source.

2. The source describes variation and does not supply evidence about that individual’s outcome; further appropriate individual clinical evidence would be required.

Made from the web lesson SCI-LAUNCH-A2-W07-L3
Version _____

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Staff mark scheme · recording

Record the raw score out of 24 and actual access or content support. For an incorrect follow-through percentage in Q5, award the comparison point only when it correctly compares the pupil's two displayed values; record this as error-carried-forward.

Do not award a [QUAL] grade. Preserve original paper responses before supported correction. These ten questions and 24 mark points are the existing checkpoint, not a second test.

Collect both learners' papers before review. Do not disclose a solution to one learner while the other is still testing. If a learner cannot finish under the agreed conditions, retain the incomplete evidence and record the reason and next arrangement. Never invent a result or fill the gap from practice.

The Supported, Independent and Aspire routes change access and later reflection, not the ten questions, marks or science help during the test.

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Staff answers · shared work

Starter

An adult may provide authorised access support or exact-word scribing, but content coaching changes what the evidence shows.

Worked example

How to answer a question without receiving content coaching.

Read or hear the question exactly.

Choose your permitted response mode.

Ask for wording to be repeated or for an agreed pause.

Keep the science answer your own; support is recorded separately.

Shared evidence task

Use the separate analytic mark scheme. Record a raw score out of 24 and actual support. No automatic [QUAL] grade or qualification claim.

Card placements

Card	Group
Read the question exactly	Access support
Allow a quiet pause	Access support
Scribe the pupil’s exact words	Access support
Tell the pupil which allele is dominant	Content coaching — record separately
Show the answer key during the test	Content coaching — record separately
Complete the Punnett square for the pupil	Content coaching — record separately

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Staff answers · Supported

Arrival

1. Readiness: Accept a suitable mode; this is an access check, not a scored science item.
2. Materials: Use only the materials authorised by the teacher for the chosen condition.
3. Help: Accept a request to read or repeat wording without content coaching.
4. Pacing: Move to another question, return later, or request an agreed pause.

Independent / application

1. Show: Mark using the separate teacher mark scheme; do not display it during testing.
2. Review: Preserve original evidence and label later supported improvements.
3. Plan: Use the item-level evidence, not an invented grade boundary.

Two-check exit

1. Locate evidence: Accept an evidence-linked choice; do not infer a score before marking.
2. Next learning: Accept a specific target linked to marked or discussed work.

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Staff answers · Independent

Arrival

1. Readiness: Accept a suitable mode; this is an access check, not a scored science item.
2. Materials: Use only the materials authorised by the teacher for the chosen condition.
3. Help: Accept a request to read or repeat wording without content coaching.
4. Pacing: Move to another question, return later, or request an agreed pause.

Independent / application

1. Show: Mark using the separate teacher mark scheme; do not display it during testing.
2. Review: Preserve original evidence and label later supported improvements.
3. Plan: Use the item-level evidence, not an invented grade boundary.

Two-check exit

1. Locate evidence: Accept an evidence-linked choice; do not infer a score before marking.
2. Next learning: Accept a specific target linked to marked or discussed work.

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Staff answers · Aspire

Arrival

1. Readiness: Accept a suitable mode; this is an access check, not a scored science item.
2. Materials: Use only the materials authorised by the teacher for the chosen condition.
3. Help: Accept a request to read or repeat wording without content coaching.
4. Pacing: Move to another question, return later, or request an agreed pause.

Independent / application

1. Show: Mark using the separate teacher mark scheme; do not display it during testing.
2. Review: Preserve original evidence and label later supported improvements.
3. Plan: Use the item-level evidence, not an invented grade boundary.

Two-check exit

1. Locate evidence: Accept an evidence-linked choice; do not infer a score before marking.
2. Next learning: Accept a specific target linked to marked or discussed work.

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Staff · source and assessment scope

Weekly outcome: Topics 2 to 3 assessment (Dec).

Exact goal: Show current understanding of the taught Autumn 2 content, then choose a precise next learning step.

Source: Launch/Autumn2_W7_2026-27/SCI_L_A2_W7L3_Topics_2_3_Assessment_Do.html; revision c4ca8ffab98e6186a9ef2fed3921644b177f0d6d.

Workbook: _authoring/science_2026-27/_toolchain/scheme_references/LAUNCH_SOW.xlsx; LAUNCH Weekly - Autumn B45 Aut2·W7; C45 above.

The classroom checkpoint samples taught Autumn 2 content. It does not assess every topic point or give a [QUAL] grade. Keep raw evidence, access arrangements and actual content support distinct.

Use fictional allele models and supplied study data. No family history, diagnosis or prediction about an individual is invited.

Keep the real first answer, precise feedback and the pupil’s next response. Marking codes require an actual event; clicks or completed boxes give no automatic R, E, Audience or Influence credit.

Spring 1 scheme lines await owner confirmation. Record any unfinished coverage and hand over the evidence without inventing the next lesson.

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